

Quiz — 1.3.2 Databases

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.3.2

Section A — Multiple choice (8 marks)

1. Compared with a flat file, a relational database typically has: A. More data redundancy B. Less data redundancy and fewer anomalies C. Only one table D. No primary keys

Your answer: ☐

2. A foreign key is: A. An attribute used only for sorting B. The primary key of another table, used to link tables C. A key made of two attributes D. A key that must always be null

Your answer: ☐

3. A composite primary key is one that: A. Links to another database B. Is made up of two or more attributes combined C. Is automatically generated D. Cannot be used to identify records

Your answer: ☐

4. A many-to-many (M:N) relationship is normally resolved by: A. Deleting one of the tables B. Adding a linking/junction table C. Removing the primary keys D. Using a secondary key only

Your answer: ☐

5. Referential integrity ensures that: A. Every foreign key references a valid existing primary key (or is null) B. All data is encrypted C. Tables can never be joined D. Records are always sorted

Your answer: ☐

6. A table is in 2NF if it is in 1NF and: A. Has no transitive dependencies B. Has no partial dependencies on a composite key C. Has no primary key D. Contains repeating groups

Your answer: ☐

7. Which ACID property guarantees that a transaction is "all or nothing"? A. Consistency B. Isolation C. Atomicity D. Durability

Your answer: ☐

8. Record locking is used during a transaction to: A. Speed up SELECT queries B. Prevent the lost update problem from concurrent edits C. Encrypt the record D. Remove transitive dependencies

Your answer: ☐

Section B — Short answer (12 marks)

9. State two differences between a flat file database and a relational database. [2]

10. Explain the rule a table must meet to be in 1NF. [2]

11. A table is in 2NF but not 3NF. Explain what must be removed to reach 3NF. [2]

12. Consider a table `Student(StudentID, Surname, Form, TutorName)`. Write an SQL query that returns the `StudentID` and `Surname` of every student in form '12A', sorted by `Surname` in ascending order. [3]

13. Explain what is meant by the Isolation property of a transaction, and why it matters. [2]

14. State one benefit and one risk of using record locking. [1]

Section C — Exam-style question (5 marks)

15. A library uses a relational database with two tables:

`Member(MemberID, Name, Email)` `Loan(LoanID, MemberID, BookTitle, DueDate)`

(a) Identify the foreign key and the table it links to. [1]

(b) Write an SQL statement that joins the two tables to list each member's `Name` together with the `BookTitle` of every book they currently have on loan. [3]

(c) Explain how referential integrity prevents an invalid loan record. [1]

Answer key

Section A (8 marks, 1 each)

1. **B** — Relational databases reduce redundancy and anomalies through normalisation and linked tables.
2. **B** — A foreign key is the primary key of another table used to create a link.
3. **B** — A composite key combines two or more attributes to uniquely identify a record.
4. **B** — A junction/linking table (with a composite key) resolves M:N.
5. **A** — Referential integrity requires foreign keys to match a valid existing primary key (or be null).
6. **B** — 2NF removes partial dependencies (non-key attributes must depend on the whole key).
7. **C** — Atomicity = all steps succeed or all are rolled back.

8. **B** — Locking a record stops concurrent edits causing a lost update.

Section B (12 marks)

9. **[2]** Any two for 1 mark each: flat file uses a single table vs multiple linked tables; flat file has high redundancy vs reduced redundancy; flat file prone to anomalies/inconsistency vs consistent; flat file simple vs scalable/complex queries.
10. **[2]** 1 mark: no repeating groups and all values are atomic (single-valued). 1 mark: the table has a (defined) primary key.
11. **[2]** 1 mark: remove transitive dependencies — where a non-key attribute depends on another non-key attribute. 1 mark: move that attribute to a separate table (so every non-key attribute depends only on the primary key).
12. **[3]** Sample SQL:

```
SELECT StudentID, Surname
FROM Student
WHERE Form = '12A'
ORDER BY Surname ASC;
```

1 mark: `SELECT StudentID, Surname FROM Student`; 1 mark: `WHERE Form = '12A'`; 1 mark: `ORDER BY Surname ASC` (ASC may be omitted as it is the default).

13. **[2]** 1 mark: concurrent transactions execute as if isolated — intermediate/uncommitted states are not visible to other transactions. 1 mark: prevents interference/inconsistent reads when transactions run at the same time.
14. **[1]** Benefit: prevents the lost update problem / keeps data consistent. Risk: can cause deadlock (or block/slow other users). (Both needed for the mark.)

Section C (5 marks)

15. (a) **[1]** `MemberID` in the `Loan` table is the foreign key; it links to `MemberID` (primary key) in the `Member` table.

(b) **[3]** Sample SQL:

```
SELECT Member.Name, Loan.BookTitle
FROM Member
JOIN Loan ON Member.MemberID = Loan.MemberID;
```

1 mark: `SELECT` of Name and BookTitle; 1 mark: `FROM / JOIN` of both tables; 1 mark: correct `ON Member.MemberID = Loan.MemberID`.

- (c) **[1]** Referential integrity prevents a `Loan` record being created with a `MemberID` that does not exist in `Member`, so there can be no orphaned loan pointing to a non-existent member.

Total: 8 + 12 + 5 = 25 marks